MARGDARSHAK

Solving the temporal-spatial coordination gap in Gwalior

TOURISM / BEYOND THE FORT

THE SPECIFIC FAILURE

Time-constrained transit travellers with 4–8 hours can lose their visit window when opening schedules, route conditions, terrain and local information are not reconciled in one itinerary.

Root cause: fragmented ownership + asynchronous schedules + unverified last-mile access data.

4–8 HOUR WINDOW | MULTI-AGENCY HOURS | STEEP / UNEVEN TERRAIN | LOCAL DATA TRUST

CURRENT FAILURE

- Maps optimise paths, not multi-authority opening conflicts.
- Generic chat can invent hours, fares or etiquette.
- Terrain and stairs are easily hidden by distance alone.

OUR INTERVENTION

Schedule filters + deterministic routing + transparent costs + source-attributed facts + verified nearby commerce.

DESIGNED FOR FOUR ACTORS

- Transit Explorer
- Heritage Seeker
- Micro-Merchant
- Data Administrator

EXTERNAL EVIDENCE [1] Gwalior District lists Gujari Mahal Museum hours as 10:00–17:00 and Monday closed. [2] Incredible India lists Gwalior Fort as 06:00–17:30.

Integrity correction: a Friday closure for Gwalior Fort was not substantiated in the sources reviewed. Do not encode it as fact until the managing authority confirms it.

A defensible problem choice beats a bigger feature list

The selection favours a failure we can test deterministically inside a 24-hour MVP.

FRAMING A — **Micro-enterprise digital invisibility**

REJECTED • Depends on merchant onboarding, consent, inventory and sustained B2B adoption; not credible to validate in 24 hours.

FRAMING B — **Temporal-spatial coordination failure**

SELECTED • Can be demonstrated with schedules, route constraints, budget guarantees and a small curated data set.

SYNERGY DECISION TRAIL / HUMAN + AI

DECISION	INITIAL VIEW	AI CONTRIBUTION	FINAL HUMAN DECISION
Problem selection	Build an artisan onboarding tool.	Flagged long-term merchant onboarding as unfeasible for a 24-hour MVP.	Selected temporal-spatial routing; measurable inputs and acceptance tests.
Technical approach	Ask an LLM to generate travel times.	Warned that generative outputs may hallucinate times and ignore elevation/access constraints.	Rejected LLM-based routing; use a deterministic Python constraint/graph solver.
Cultural norms	Let AI generate site dress codes.	Flagged risk of fabricating etiquette for religious and heritage sites.	Display only pre-verified, source-attributed rules approved by a human curator.

CASE-FILE INTEGRITY The “67% concentration” figure is treated as a scenario signal supplied in the case file, not as independently verified tourism research.

SOLUTION ARCHITECTURE

Hybrid engine: AI interprets; rules decide

One strict boundary: AI never determines whether a route is open, safe, affordable or feasible.

01 INPUT

Traveller parameters

Time: 2–10 h
Origin: Gwalior Junction by default
Day/time + interests
“Avoid steep stairs” toggle

02 PARSE + CONFIRM

NLP / Hinglish parser

Convert natural language into a structured request.
User confirms parsed date, time, duration, interests and mobility needs.

03 RULE ENGINE

Deterministic solver

Filter opening windows.
Optimise graph route + access flags.
Calculate authority tariffs.
Guarantee total time ≤ budget.

04 OUTPUT

Explainable itinerary

Arrival / visit / departure times
Route + transit mode
Itemised agency costs
Verified merchant + cited notes

GROUNDED DATA + HUMAN CONTROL

VERIFIED LOCAL DATA

JSON schema + local SQLite cache: hours, authority, tariffs, map nodes, access flags, source URL and last-verified date.

AI / RAG ONLY

Parse intent and summarise retrieved historical text only from the approved document store. No open-ended fact generation.

HUMAN-IN-THE-LOOP

Curators verify hours, tariffs, source texts and merchant willingness. Users confirm parsed parameters before routing.

ACTORS

- Transit Explorer (4–8 h)
- Cultural Heritage Seeker
- Local Micro-Merchant
- Data Administrator

NON-FUNCTIONAL TARGETS

< 1.5 s generation • offline on cached data • responsive to 360 px • no registration / background location tracking • deterministic fallback if LLM is unavailable

VERIFICATION

Acceptance tests + requirements + guardrails

Demo expected behaviour with seeded fixtures; production data is not asserted until source and field verification.

ACCEPTANCE TESTS

TEST 01 — Temporal feasibility

Monday 10:30 + 5 h + Gujari Mahal → flag Monday closure, cite [1], then suggest only alternatives with verified hours.

TEST 02 — Topological / mobility constraint

“Avoid steep stairs” + Fort → use Urwahi/Urwai vehicle-ramp candidate, exclude stair segment in demo graph; field-verify accessibility [3].

TEST 03 — Micro-merchant inclusion

4 h architecture tour → fixture adds a curator-approved merchant within 800 m; no verified match means no invented recommendation.

ERROR HANDLING

- Data older than 90 days → “Requires Local Confirmation”.
- Under 2.5 h + Bateshwar → reject infeasible trip; offer a city alternative.
- LLM unavailable → deterministic itinerary still works.

FUNCTIONAL REQUIREMENTS / FR-1 TO FR-8

FR-1	Filter closed attractions by day/time.
FR-2	Optimise route; reduce backtracking and impassable stairs.
FR-3	Enforce total itinerary time ≤ user budget.
FR-4	Show transparent tariffs by managing authority.
FR-5	Include verified merchant within 800 m of each major stop.
FR-6	Ground facts in approved documents and citations.
FR-7	Respect “Avoid Steep Stairs” mobility toggle.
FR-8	Show authority + last-verified date for each key data item.

SCHEDULE INTEGRITY: Gujari Mahal Monday closure is sourced [1]. A “Fort closed Friday” rule is unverified and must not be hard-coded.

NFR / OPERATING TARGETS

<1.5 s • cached/offline mode • 360 px responsive • no login/tracking • LLM fallback

BUILD ROUND PLAN

A two-week path from constraint proof to tested prototype

Build the smallest auditable engine first; add retrieval only after the deterministic path is stable.

PHASE 1 / CURRENT MVP

Python mock constraint engine
JSON schema + test fixtures
Excalidraw wireframes
Goal: prove deterministic logic

PHASE 2 / WEEK 1

Streamlit or React front end
Local SQLite for 20 core nodes
Wire schedule + access metadata
Goal: runnable local prototype

PHASE 3 / WEEK 2

Lightweight RAG for curated history
Source-cited summaries only
Test with 5 transit travellers
Goal: validate clarity and fit

V1 WILL NOT DO

× Live ticketing / payments
× Real-time GPS turn-by-turn voice navigation
× Open merchant self-registration
× Open-ended historical generation
× Morena / Chambal peripheral circuit in a 4–8 h itinerary

INTEGRITY + OWNERSHIP

• Case-file figures (including 67%) are scenario signals, not independent findings.
• Team must explain why routing is deterministic and AI is bounded.
• Asset disclosure: Excalidraw, Google Colab, Qwen AI used for wireframes, mock code and framing/SRS drafting.
• AI interaction log is referenced in the Brief; add its share link to the submitted package.

EXTERNAL EVIDENCE

[1] District Gwalior - Tourist Places

Gujari Mahal: 10:00–17:00; closed Mondays.

[2] Incredible India - Gwalior Fort

Published hours: 06:00–17:30.

[3] Incredible India - Gwalior Travel Guide

Describes Qila Gate for pedestrians and Urwahi Gate vehicle access via ramp.

SUBMISSION DECLARATION

All real-world hours, access claims, tariffs and merchant listings must be source-attributed and carry a last-verified date. Stale data (>90 days) is visibly flagged. The current MVP demonstrates logic with curated fixtures; it does not claim live, complete city coverage.